

Shared Risk-Capital Constraints in Dynamic Option Dealer Hedging

Yuvraj Malik, Nicholas Bo You Chin

Correspondence: u1547055@utah.edu, u1552776@utah.edu

Abstract

We develop a theory-first model of dynamic option dealer hedging when several dealers draw on the same limited risk-capital or balance-sheet resource. Dealers hold option books, choose hedge trades in the underlying, and face intertemporal hedge-inventory and trade-adjustment costs. A common internal capacity constraint couples their feasible hedge choices. We formulate the problem as a shared-constraint generalized Nash game and study its variational equilibrium, which assigns a single shadow cost to the shared resource. In a quadratic benchmark, the equilibrium is equivalent to a finite-dimensional variational inequality and to the minimizer of an exact convex potential. We prove existence, uniqueness under private regularization, multiplier comparative statics, sensitivity to inventory shocks, and an envelope formula linking the common multiplier to contract-level premia. A pricing rule based on marginal equilibrium cost yields a first-order implied-volatility wedge. Structural simulations illustrate how tighter shared capacity suppresses hedging and produces larger wedges for contracts with higher marginal risk-capital intensity.

Keywords. option hedging; dealer inventory; generalized Nash equilibrium; variational inequality; funding liquidity; balance-sheet constraints; implied volatility

MSC codes. 91G20; 91G80; 91A10; 90C33; 49J40

1 Introduction

Dealer hedging links option markets to underlying-market liquidity. In a frictionless complete-market benchmark, a dealer's option inventory can be treated as a single-agent hedging problem. The dealer trades the underlying asset and a cash account to replicate, super-replicate, or approximately hedge the book. This benchmark is useful, but it suppresses an important feature of actual dealer markets: hedging capacity is not unlimited and is not privately owned in isolation. Dealers face margin requirements, inventory limits, internal risk limits, funding costs, and finite market depth. When several dealers hedge option books at the same time, their feasible hedge choices may be coupled by a common scarcity of funding, liquidity, or balance-sheet capacity.

This paper develops a theory-first model of that coupling. The key modeling step is to treat shared balance-sheet capacity as a common risk-capital constraint in a multi-dealer hedge game. Each dealer chooses hedge trades against an option inventory, but the feasible set contains a market-level capacity constraint. Because the common constraint depends on all dealers' decisions, the model is a generalized Nash equilibrium problem. We focus on the variational equilibrium, the equilibrium selection associated with a single multiplier on the shared constraint. That multiplier has a direct financial interpretation: it is the endogenous shadow cost of scarce funding or balance sheet.

The model is dynamic in the finite-horizon sense. Dealers carry hedge positions through time. Current trades change future hedge inventory, and trades are penalized for rapid adjustment through a quadratic smoothing cost. Hence the pseudo-gradient is no longer a collection of independent static stage games. It has intertemporal structure generated by cumulative hedge inventory and adjacent-trade costs. The time index therefore represents a finite-horizon dynamic hedge program rather than repeated unrelated one-period games.

The contribution is deliberately theoretical. We do not claim to estimate dealer inventories or validate the model empirically. Instead, we isolate a tractable mechanism and derive its equilibrium implications. In the benchmark model, the variational equilibrium is characterized by a finite-dimensional variational inequality. The same equilibrium is also the minimizer of an exact convex potential over the coupled feasible set. This structure yields existence, uniqueness under mild private regularization, multiplier comparative statics with respect to capacity, stability bounds for inventory shocks, and an envelope formula connecting

the common multiplier to contract-level premia. A marginal pricing rule then maps the premium into a first-order implied-volatility wedge through option vega.

The paper addresses a gap between three established literatures. The option-market literature has shown that dealer demand, inventory, and hedging pressure can affect option prices and implied volatility. The funding-liquidity and intermediary-asset-pricing literatures have shown that scarce balance sheet produces shadow costs and pricing wedges. The variational-inequality and generalized-Nash literatures provide tools for shared-constraint equilibria. What is missing is a compact mathematical finance model in which multiple option dealers hedge dynamically under a shared capacity constraint, the equilibrium is represented as a variational inequality, and the common multiplier is mapped into option-level premia.

The rest of the paper is organized as follows. Section 2 positions the paper relative to prior work. Section 3 presents the dynamic dealer environment, the shared risk-capital constraint, and the variational-equilibrium formulation. Section 4 gives the main theoretical results. Section 5 presents structural numerical illustrations. Section 6 discusses interpretation and limitations. Section 7 concludes. Proofs are collected in the Appendix.

2 Related literature and positioning

The first related strand studies option demand, dealer inventory, and implied-volatility distortions. Bollen and Whaley [3] show that net buying pressure affects implied volatility. Garleanu, Pedersen, and Poteshman [15] develop a demand-based option-pricing model in which end-user demand and dealer risk-bearing capacity generate option price deviations. Lakonishok et al. [21] document systematic option-market activity across investor classes. Ni, Pearson, and Poteshman [22] link option markets and underlying prices through hedging-related effects around expiration. Recent empirical work studies how gamma positioning and dealer constraints affect market quality and funding-illiquidity measures inferred from derivatives [6, 16]. These papers motivate the financial mechanism, but they do not formulate dealer hedging as a shared-constraint generalized Nash game with a variational-equilibrium multiplier.

A second strand studies funding liquidity, intermediary balance sheets, and margin-based pricing. Brunnermeier and Pedersen [5] show how market liquidity and funding liquidity reinforce each other. Garleanu and Pedersen [14] model deviations from the law of one price caused by margin requirements and shadow costs of capital. He and Krishnamurthy [19] and Adrian, Etula, and Muir [1] emphasize the asset-pricing role of intermediary balance sheets. This paper takes the shadow-cost logic from that literature and embeds it into a tractable multi-dealer option-hedging equilibrium. The common multiplier in our model is the analogue of a balance-sheet shadow cost, but it is generated endogenously by a shared hedge-capacity constraint.

A third strand studies hedging with frictions, feedback, and constraints. Dynamic hedging can feed back into prices and volatility, as in Frey and Stremme [13] and Sircar and Papanicolaou [26]. Large-trader and illiquidity models show that hedging itself can change prices [8, 12]. Gamma-constrained super-replication and hedging under market impact are developed by Sonar and Touzi [27] and Bouchard, Loeper, and Zou [4]. These models are typically single-agent or representative-agent models. Our paper keeps the frictional-hedging motivation but shifts the object to a multi-dealer shared-constraint equilibrium.

A fourth strand studies market making, execution, and strategic interaction under impact. Avellaneda and Stoikov [2], Stoikov and Saglam [28], and Gueant, Lehalle, and Fernandez-Tapia [17] develop inventory-based market-making models. Strategic liquidity and impact games appear in Carlin, Lobo, and Viswanathan [7] and Schied and Zhang [25]. These works show why simultaneous trading decisions cannot always be decomposed into independent controls. Our model is closer to this equilibrium view, but the coupling is a shared balance-sheet capacity constraint tied to option-book hedge pressure rather than only transient execution impact.

The mathematical framework comes from variational inequalities and generalized Nash equilibrium problems. Foundational results on Nash equilibrium, variational inequalities, and generalized Nash games are given by Rosen [24], Facchinei and Pang [9], Harker [18], Facchinei, Fischer, and Piccialli [10], and Facchinei and Kanzow [11]. The variational equilibrium is a refinement of generalized Nash equilibrium for shared constraints and has been studied by Kulkarni and Shanbhag [20]. We use this framework because it gives a single multiplier for the common capacity constraint. That single multiplier is central to the financial interpretation.

Relative to these literatures, the paper contributes a minimal theory model with four linked features:

dynamic hedge inventory, multiple dealers, shared risk-capital capacity, and an option-level premium decomposition through the variational-equilibrium multiplier. The point is not to reproduce all institutional detail. The point is to make one equilibrium mechanism mathematically explicit.

3 Model

3.1 Dealer books and hedge inventory

Time is discrete: $t = 0, 1, \dots, T - 1$. There are N dealers. Dealer i holds an option inventory $q_{i,k,t}$ in contracts $k = 1, \dots, K$. At date t , the underlying price is S_t , the one-period volatility input is σ_t , and the time increment is Δt . Let $\Delta_{k,t}$, $\Gamma_{k,t}$, and $\mathcal{V}_{k,t}$ denote, respectively, the delta, gamma, and vega of contract k at date t under a reference pricing model. These Greeks are used to construct state variables and risk-capital charges; the equilibrium model is not a Black-Scholes replication argument.

Dealer i controls trades in the underlying. Let $h_{i,t}$ denote dealer i 's hedge position in shares before trading at date t and let $u_{i,t}$ be the trade in shares at date t . The hedge position evolves as

$$h_{i,t+1} = h_{i,t} + u_{i,t}, \quad h_{i,0} \text{ given.} \quad (3.1)$$

Equivalently, the post-trade hedge position at date t is $h_{i,0} + \sum_{s=0}^t u_{i,s}$. Thus current trades affect future hedge positions. This is the first source of intertemporal coupling.

The option book delta exposure, measured in shares of the underlying, is

$$\Delta_{i,t}^{\text{book}} := \sum_{k=1}^K q_{i,k,t} \Delta_{k,t}. \quad (3.2)$$

A dealer who is long positive delta must short the underlying to delta hedge. Hence the target hedge position is

$$h_{i,t}^{\text{tar}} := -\Delta_{i,t}^{\text{book}}. \quad (3.3)$$

The sign convention in (3.3) is important: the hedge target is the underlying position that offsets the book delta, not the book delta itself. The post-trade hedge error is

$$e_{i,t}(u_i) := h_{i,t} + u_{i,t} - h_{i,t}^{\text{tar}}. \quad (3.4)$$

3.2 Dealer objective

Dealer i chooses the trade sequence $u_i = (u_{i,0}, \dots, u_{i,T-1}) \in \mathbb{R}^T$. Let $U_t = \sum_{j=1}^N u_{j,t}$ denote aggregate dealer trading at date t . Dealer i minimizes

$$\begin{aligned} J_i(u_i, u_{-i}) := & \sum_{t=0}^{T-1} \left[\frac{\rho_i}{2} (h_{i,t} + u_{i,t} - h_{i,t}^{\text{tar}})^2 + \frac{\eta_i}{2} u_{i,t}^2 + \frac{\chi_i}{2} (u_{i,t} - u_{i,t-1})^2 \right. \\ & \left. + \kappa \left(\frac{1}{2} u_{i,t}^2 + u_{i,t} \sum_{j \neq i} u_{j,t} \right) \right], \end{aligned} \quad (3.5)$$

where $u_{i,-1}$ is given. The first term penalizes hedge shortfall. The second term is a private trading or execution penalty. The third term is an inventory-adjustment or trade-smoothing cost. It captures the fact that aggressive changes in hedging speed consume execution resources and create liquidity costs. This term makes the problem genuinely dynamic because the marginal cost of $u_{i,t}$ depends on adjacent trades.

The last term is a Cournot-style aggregate impact term. Dealer i internalizes its own trading cost and the effect of its trade against the aggregate flow of other dealers. Its derivative is

$$\frac{\partial}{\partial u_{i,t}} \kappa \left(\frac{1}{2} u_{i,t}^2 + u_{i,t} \sum_{j \neq i} u_{j,t} \right) = \kappa U_t. \quad (3.6)$$

Thus the pseudo-gradient contains an aggregate-pressure term without mechanically charging every dealer the entire aggregate social cost $\kappa U_t^2/2$ as a private cost. The associated potential contains the aggregate term $\kappa U_t^2/2$.

3.3 Greek-based internal risk-capital usage

The shared constraint is an internal risk-capital constraint, not a literal exchange margin formula. It represents a balance-sheet or risk-capital budget allocated across dealers by a common funding desk, central clearing constraint, or firm-wide risk system.

For dealer i , define the post-hedge residual dollar delta exposure as

$$D_{i,t}(u_i) := S_t \left(h_{i,t} + u_{i,t} + \sum_{k=1}^K q_{i,k,t} \Delta_{k,t} \right). \quad (3.7)$$

Since $h_{i,t}^{\text{tar}} = -\sum_k q_{i,k,t} \Delta_{k,t}$, this is $S_t e_{i,t}(u_i)$. Define the dollar gamma of the option book by

$$G_{i,t} := S_t^2 \sum_{k=1}^K q_{i,k,t} \Gamma_{k,t}. \quad (3.8)$$

To motivate the risk-capital charge, consider a one-period return shock

$$\frac{\Delta S_t}{S_t} = \sigma_t \sqrt{\Delta t} Z_{t+1}, \quad Z_{t+1} \sim N(0, 1).$$

A second-order approximation to the marked P&L of the residual hedged book is

$$\Delta V_{i,t} \approx D_{i,t}(u_i) \sigma_t \sqrt{\Delta t} Z_{t+1} + \frac{1}{2} G_{i,t} \sigma_t^2 \Delta t Z_{t+1}^2. \quad (3.9)$$

This approximation motivates a quadratic internal capital charge based on local Greek P&L risk:

$$r_{i,t}(u_i) := \beta_D D_{i,t}(u_i)^2 \sigma_t^2 \Delta t + \frac{\beta_G}{2} G_{i,t}^2 \sigma_t^4 \Delta t^2, \quad (3.10)$$

where $\beta_D, \beta_G \geq 0$ are capital weights. The first term charges residual dollar delta risk; the second term charges dollar gamma risk. The dollar-gamma term depends on the option book and not on the hedge trade. It therefore tightens the date- t capacity constraint as a baseline risk-capital use, but it does not enter the hedge first-order condition directly. It enters contract premia through the envelope term because a marginal contract changes $G_{i,t}$ and hence changes the available capacity for all dealers. The exact constants are not essential for the equilibrium mechanism. What matters is that the charge has an explicit Greek-risk primitive and is convex in the hedge decision through the residual delta term.

The full balance-sheet usage of dealer i is

$$m_{i,t}(u_i) := \alpha_i u_{i,t}^2 + r_{i,t}(u_i), \quad \alpha_i > 0. \quad (3.11)$$

The term $\alpha_i u_{i,t}^2$ is an internal capital charge for hedge financing or execution balance-sheet usage. It can be interpreted as a Markowitz-style quadratic allocation rule: if a funding desk charges dealers according to the conditional variance of hedge-financing demand, then a quadratic charge in hedge size is natural. This choice is analytically convenient, but it is not the only possible specification. A linear haircut-type constraint such as $\sum_i \alpha_i |u_{i,t}| \leq M_t$ leads to a nonsmooth variational inequality with subgradient conditions. The present paper focuses on the smooth quadratic benchmark because it exposes the role of the common multiplier while preserving a tractable potential structure.

The shared capacity constraint is

$$\sum_{i=1}^N m_{i,t}(u_i) \leq M_t, \quad t = 0, \dots, T-1, \quad (3.12)$$

where $M_t > 0$ is the available internal risk-capital or balance-sheet capacity at date t . Define the feasible set

$$\mathcal{K} := \left\{ u \in \mathbb{R}^{NT} : \sum_{i=1}^N m_{i,t}(u_i) \leq M_t \text{ for all } t \right\}. \quad (3.13)$$

Additional execution bounds may be included, but they are not needed for compactness when $\alpha_i > 0$ and $M_t < \infty$.

Assumption 3.1 (Standing assumptions). The parameters satisfy $\rho_i, \eta_i, \chi_i, \kappa \geq 0$, $\alpha_i > 0$, and $\beta_D, \beta_G \geq 0$. The capacity levels satisfy $M_t > 0$, and the feasible set $\mathcal{K}$ is nonempty. Slater's condition holds whenever KKT multipliers are invoked.

3.4 Generalized Nash equilibrium and variational equilibrium

The shared constraint (3.12) couples the feasible decisions of all dealers. Hence the model is a generalized Nash equilibrium problem rather than an ordinary Nash game.

Definition 3.2 (Generalized Nash equilibrium). A feasible profile $u^* \in \mathcal{K}$ is a generalized Nash equilibrium if, for every dealer i ,

$$u_i^* \in \operatorname{argmin}_{v_i} \{ J_i(v_i, u_{-i}^*) : (v_i, u_{-i}^*) \in \mathcal{K} \}.$$

We focus on the variational equilibrium selection. This selection imposes a common multiplier on the shared constraint at each date. Economically, the variational selection is appropriate when the shared constraint represents a common funding desk, a central clearing constraint, a firm-wide balance-sheet allocation, or a market-wide funding resource faced symmetrically by dealers. It is more restrictive than a fully heterogeneous generalized Nash equilibrium. If dealers face different prime brokers, different bilateral funding rates, or dealer-specific balance-sheet constraints, then the appropriate model would allow dealer-specific multipliers. The variational equilibrium should therefore be read as a benchmark for common scarcity, not as a universal description of every dealer market.

Let $F : \mathbb{R}^{NT} \rightarrow \mathbb{R}^{NT}$ be the pseudo-gradient obtained by stacking player gradients:

$$F(u) := (\nabla_{u_1} J_1(u), \dots, \nabla_{u_N} J_N(u)). \quad (3.14)$$

Definition 3.3 (Variational equilibrium). A vector $u^* \in \mathcal{K}$ is a variational equilibrium if it solves

$$\langle F(u^*), u - u^* \rangle \geq 0, \quad \forall u \in \mathcal{K}. \quad (3.15)$$

Because hedge inventory and adjustment costs couple decisions across dates, the map F is not block diagonal in time. It is affine with intertemporal structure from cumulative hedge inventory and the adjacent-trade term $(u_{i,t} - u_{i,t-1})^2$.

4 Main results

4.1 Potential representation

The quadratic benchmark admits an exact potential. Define

$$\mathcal{P}(u) := \sum_{i=1}^N \sum_{t=0}^{T-1} \left[\frac{\rho_i}{2} (h_{i,t} + u_{i,t} - h_{i,t}^{\text{tar}})^2 + \frac{\eta_i}{2} u_{i,t}^2 + \frac{\chi_i}{2} (u_{i,t} - u_{i,t-1})^2 \right] + \frac{\kappa}{2} \sum_{t=0}^{T-1} U_t^2. \quad (4.1)$$

Proposition 4.1 (Exact potential). *For the game defined by (3.5), the pseudo-gradient satisfies $F = \nabla \mathcal{P}$.*

Proposition 4.2 (Variational equilibrium and potential minimization). *Under Assumption 3.1, if $\mathcal{P}$ is convex, then u^* is a variational equilibrium if and only if u^* solves*

$$\min_{u \in \mathcal{K}} \mathcal{P}(u). \quad (4.2)$$

If $\mathcal{P}$ is strongly convex on $\mathcal{K}$, the variational equilibrium is unique.

The potential representation clarifies the role of the impact term. Individual dealers face the Cournot-style marginal aggregate pressure κU_t , while the potential aggregates this pressure as $\kappa U_t^2 / 2$.

4.2 Existence, uniqueness, and regularity

The capacity constraint compactifies feasible trade sequences because $\alpha_i u_{i,t}^2 \leq m_{i,t}(u_i) \leq M_t$. Therefore an additional box constraint is not needed for existence.

Theorem 4.3 (Existence). *Under Assumption 3.1, if $\mathcal{P}$ is convex, the variational equilibrium exists.*

Theorem 4.4 (Uniqueness under private regularization). *Suppose Assumption 3.1 holds and $\eta_i > 0$ for all dealers. Then $\mathcal{P}$ is strongly convex, and the variational equilibrium is unique.*

The condition $\eta_i > 0$ is sufficient, not necessary. Positive adjustment costs $\chi_i > 0$ also improve conditioning by penalizing high-frequency changes in hedge trades, but a positive private level cost is the cleanest assumption for uniqueness in every dealer-time direction.

4.3 KKT system and common funding multiplier

Under Slater's condition, the variational equilibrium satisfies the KKT system for the convex program (4.2). There exist multipliers $\lambda_t \geq 0$, one for each shared capacity constraint, such that

$$0 = \nabla_u \mathcal{P}(u^*) + \sum_{t=0}^{T-1} \lambda_t \nabla_u \left(\sum_{i=1}^N m_{i,t}(u_i^*) - M_t \right), \quad (4.3)$$

with complementarity conditions

$$\lambda_t \geq 0, \quad \sum_{i=1}^N m_{i,t}(u_i^*) \leq M_t, \quad \lambda_t \left(\sum_{i=1}^N m_{i,t}(u_i^*) - M_t \right) = 0. \quad (4.4)$$

The multiplier λ_t is the shadow value of one additional unit of shared risk-capital capacity at date t . When the constraint is slack, $\lambda_t = 0$. When it binds, $\lambda_t > 0$ raises the effective marginal cost of trades and residual exposures that consume the shared resource.

4.4 Capacity comparative statics

Let

$$V(M) := \min_{u \in \mathcal{K}(M)} \mathcal{P}(u)$$

denote the optimized potential as a function of the capacity path $M = (M_0, \dots, M_{T-1})$. Under differentiability and a locally stable active set,

$$\frac{\partial V}{\partial M_t} = -\lambda_t. \quad (4.5)$$

Thus λ_t is the marginal reduction in equilibrium potential cost from relaxing the date- t capacity constraint.

The comparative-static mechanism is clearest in a one-date local representation. This representation must use the curvature of the actual capacity constraint, not only the hedge-financing coefficient α_i . Around one date, write the binding capacity usage as

$$q(u) = u^\top Q u + 2a^\top u + c_0, \quad (4.6)$$

where $Q \succeq 0$ is the effective Hessian of capacity usage with respect to hedge trades, a collects linear terms from residual delta exposure, and c_0 contains terms independent of trades. In the one-date diagonal case implied by (3.11),

$$Q_t = \text{diag}(\alpha_i + \beta_D S_t^2 \sigma_t^2 \Delta t)_{i=1}^N, \quad (4.7)$$

while the dollar-gamma charge contributes to c_0 rather than to Q_t . In the full finite-horizon model, residual delta depends on cumulative hedge inventory, so Q is the corresponding positive semidefinite intertemporal capacity Hessian.

Let the local potential have stationarity $Au - b = 0$ before the capacity multiplier is imposed, with $A \succ 0$. On a binding branch, the KKT stationarity condition is

$$Au(\lambda) - b + 2\lambda(Qu(\lambda) + a) = 0. \quad (4.8)$$

The capacity usage along this branch is $\varphi(\lambda) = q(u(\lambda))$. Differentiating (4.8) gives

$$\frac{du}{d\lambda} = -2(A + 2\lambda Q)^{-1}(Qu(\lambda) + a).$$

Therefore

$$\varphi'(\lambda) = -4(Qu(\lambda) + a)^\top (A + 2\lambda Q)^{-1} (Qu(\lambda) + a) \leq 0, \quad (4.9)$$

with strict inequality whenever $Qu(\lambda) + a \neq 0$. Hence lower capacity requires a weakly higher multiplier on the binding branch, and a strictly higher multiplier whenever the constraint has nonzero local marginal exposure. This is the formal version of the funding-scarcity mechanism.

Proposition 4.5 (Capacity monotonicity on a binding branch). *Consider the local quadratic representation (4.6)–(4.8) with $A \succ 0$ and $Q \succeq 0$. On a strictly binding branch with stable active set, the equilibrium multiplier is nonincreasing in capacity. If $Qu(\lambda) + a \neq 0$, tightening capacity strictly raises the multiplier.*

4.5 A homogeneous one-date comparative static

The aggregate-pressure term has a particularly transparent effect in a homogeneous one-date special case. This paragraph is not used as a claim about the full heterogeneous dynamic model with dealer-specific (ρ_i, η_i, χ_i) and intertemporal adjustment costs. In the full model, the corresponding linear operator is the block-tridiagonal Hessian of the dynamic potential, and the same type of sensitivity analysis can be carried out through that Hessian. The closed-form decomposition below is included only to isolate the economic role of the aggregate-pressure coefficient κ .

Consider a single date, no adjustment-cost term, no active capacity multiplier, and homogeneous private coefficients (ρ, η) . In this one-date benchmark, the pseudo-gradient matrix is

$$A = (\rho + \eta)I + \kappa \mathbf{1}\mathbf{1}^\top. \quad (4.10)$$

Its eigenvalue along the common direction $\mathbf{1}$ is $\rho + \eta + \kappa N$, while its eigenvalue on $\{\mathbf{1}\}^\perp$ is $\rho + \eta$. If a target shock decomposes as $d = d^\parallel + d^\perp$, with $d^\parallel \in \text{span}\{\mathbf{1}\}$ and $d^\perp \perp \mathbf{1}$, then the unconstrained one-date response satisfies

$$\|u^\parallel\| \leq \frac{\rho}{\rho + \eta + \kappa N} \|d^\parallel\|, \quad \|u^\perp\| \leq \frac{\rho}{\rho + \eta} \|d^\perp\|. \quad (4.11)$$

Thus, in this homogeneous static special case, aggregate-pressure costs damp crowded, common-direction hedge demand more strongly than offsetting cross-sectional hedge demand. In the heterogeneous dynamic model, this intuition survives through the spectrum of the dynamic Hessian, but the simple two-eigenvalue formula (4.11) should not be read as a closed-form characterization of the full finite-horizon equilibrium.

4.6 Contract-level premia and implied-volatility wedges

The equilibrium model produces hedge choices and multipliers. To connect these objects to option prices, we specify a dealer pricing rule. Let q_ℓ denote the net dealer inventory of contract ℓ . Let

$$V(q) := \min_{u \in \mathcal{K}(q)} \mathcal{P}(u; q) \quad (4.12)$$

denote the optimized potential as a function of the inventory vector. We define the dealer's marginal required premium for absorbing one additional unit of contract ℓ as

$$\pi_\ell(q) := \frac{\partial V(q)}{\partial q_\ell}, \quad (4.13)$$

whenever the derivative exists. If V is not differentiable, $\pi_\ell(q)$ is understood as an element of the subdifferential $\partial_{q_\ell} V(q)$. This pricing rule says that dealers require compensation equal to the marginal increase in equilibrium hedging, adjustment, impact, and balance-sheet costs generated by the additional contract.

Under differentiability of the primitives and strict complementarity of the active capacity constraints, the envelope theorem gives

$$\pi_\ell(q) = \frac{\partial \mathcal{P}(u^*(q); q)}{\partial q_\ell} + \sum_{t=0}^{T-1} \lambda_t^*(q) \frac{\partial}{\partial q_\ell} \left(\sum_{i=1}^N m_{i,t}(u_i^*(q); q) - M_t \right). \quad (4.14)$$

The first term captures the direct effect of the additional option on hedge mismatch and adjustment costs. The second term captures the balance-sheet channel. It is large when the contract increases residual Greek risk or hedge-financing usage in states where the shared constraint binds.

Equation (4.14) is the formal link between the common funding multiplier and contract-level premia. The multiplier alone does not produce an option wedge; the wedge is produced by the product of the multiplier and the contract's marginal use of scarce risk capital.

To translate the premium into an implied-volatility movement, let $C_\ell(\sigma)$ be the reference option price as a function of implied volatility and let $\mathcal{V}_\ell = \partial C_\ell / \partial \sigma$ be its vega. If the required premium π_ℓ is small relative to the option price scale, then a first-order Taylor expansion gives

$$C_\ell(\sigma_\ell + \Delta\sigma_\ell) - C_\ell(\sigma_\ell) = \mathcal{V}_\ell \Delta\sigma_\ell + o(\Delta\sigma_\ell). \quad (4.15)$$

Setting the price change equal to the required premium yields

$$\Delta\sigma_\ell = \frac{\pi_\ell(q)}{\mathcal{V}_\ell} + o(\pi_\ell(q)). \quad (4.16)$$

Thus implied-volatility wedges are larger for contracts with high marginal equilibrium cost per unit vega. Short-dated, high-gamma, or balance-sheet-intensive contracts can display larger implied-volatility distortions when the shared capacity constraint binds.

Proposition 4.6 (Premium and implied-volatility wedge). *Assume that $V(q)$ is differentiable at q , strict complementarity holds for the active shared constraints, and contract ℓ has positive vega $\mathcal{V}_\ell > 0$. Then the marginal dealer premium for contract ℓ is given by (4.14), and the corresponding first-order implied-volatility wedge is given by (4.16). If differentiability fails, the same statement holds with $\pi_\ell(q) \in \partial_{q_\ell} V(q)$ and the implied-volatility wedge is set-valued to first order.*

5 Structural numerical illustrations

The simulations in this section are not empirical tests. They are structural numerical illustrations of the theoretical model. Their purpose is to show how heterogeneous dealer option books map into hedge targets, capacity usage, the variational-equilibrium multiplier, and contract-level wedges.

5.1 Simulation design

We simulate $N = 8$ dealers over $T = 80$ time steps and $K = 45$ European options on a strike-maturity grid. The larger heterogeneous panel is used because the revised model is genuinely dynamic: the decision vector has $NT = 640$ hedge trades, cumulative hedge inventory couples dates, and the capacity constraints create T common multipliers. Dealer inventories are heterogeneous across strikes, maturities, signs, and magnitudes. At each time step, the inventory matrix generates book deltas, book gammas, and Greek-risk charges through (3.7)–(3.10). The equilibrium is computed by solving the convex potential program (4.2). The multiplier vector is computed from the active KKT stationarity system after convergence; equivalently, it is the optimal dual variable associated with the date-by-date capacity constraints, up to numerical solver tolerance.

Table 5.1: Parameters for the structural simulation

Quantity	Value	Interpretation
Dealers N	8	Heterogeneous dealers sharing a common risk-capital resource
Time steps T	80	Finite-horizon hedge program
Contracts K	45	Strike-maturity grid of European options
ρ_i	around 1.1	Penalty on post-trade hedge shortfall
η_i	around 0.4	Private trading-cost / regularization parameter
χ_i	around 1.0	Trade-adjustment cost
κ	0.22	Aggregate hedge-pressure interaction strength
α_i	0.020–0.030	Dealer-specific hedge-financing capital weights
Spot S_0	100	Reference underlying level
Base volatility	24%	Input for option Greeks
Capacity path	time-varying	Loose outside stress dates, binding near peak hedge demand

5.2 Multiplier and hedge shortfall

Figure 5.1 shows capacity, unconstrained risk-capital usage, constrained equilibrium usage, and the common multiplier. The multiplier is zero when the unconstrained hedge fits inside the capacity budget. When capacity falls below unconstrained usage, the constraint binds, equilibrium usage lies on the boundary, and the multiplier rises.

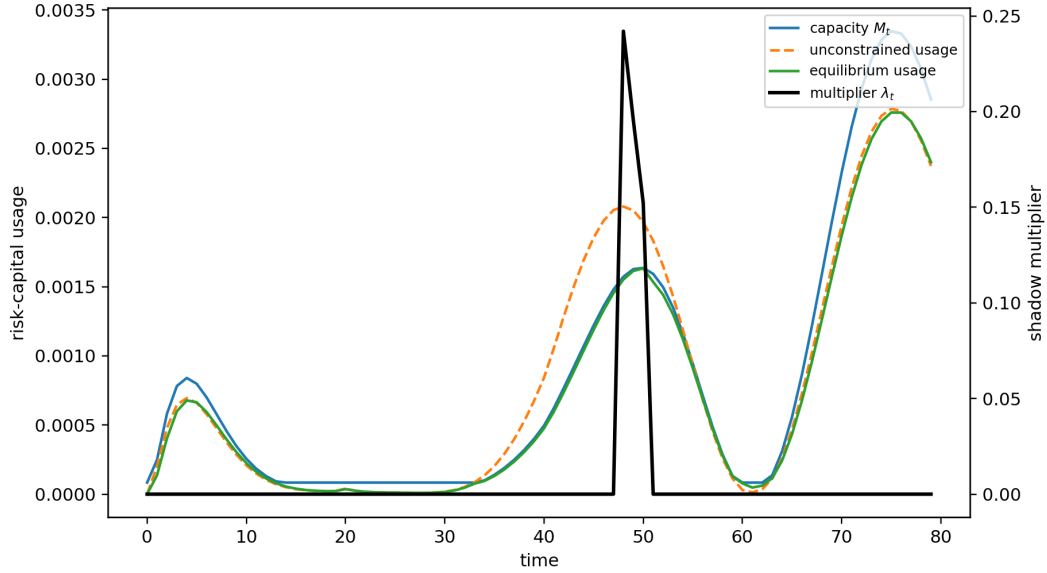


Figure 5.1: Risk-capital capacity, unconstrained usage, equilibrium usage, and the common multiplier. The multiplier is positive exactly when the shared constraint binds.

Figure 5.2 gives a dealer-level view at the peak stress date. The constrained equilibrium hedge differs from the unconstrained hedge because the common resource makes hedge demand costly at the margin. The allocation depends on target hedge demand, dealer-specific capital weights, and aggregate-pressure costs.

Figure 5.3 plots hedge shortfall and the trade-adjustment norm against the multiplier. The shortfall rises when the multiplier rises because the common capacity constraint pushes trades away from their unconstrained hedge targets. The adjustment norm shows that the dynamic smoothing term also affects the hedge path, not just the one-date allocation.

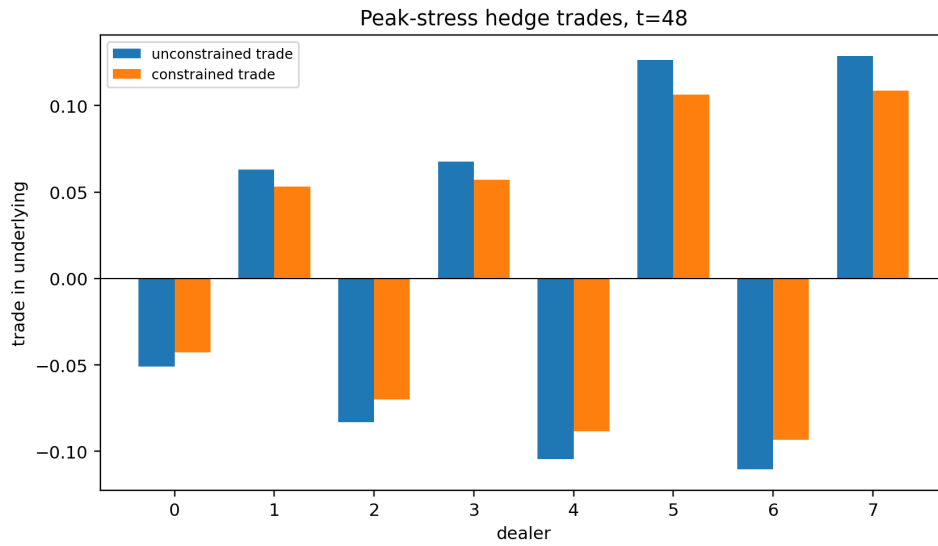


Figure 5.2: Dealer-level unconstrained and constrained hedge trades at the peak stress date. Shared capacity changes the cross-sectional hedge allocation across dealers.

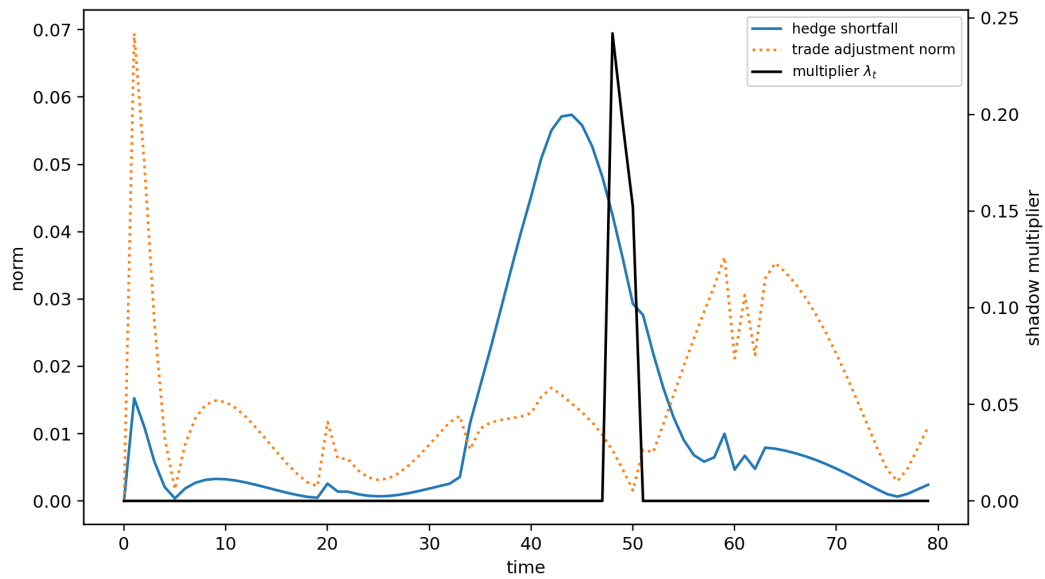


Figure 5.3: Hedge shortfall, trade adjustment, and common multiplier. Binding capacity suppresses hedge trades relative to the unconstrained benchmark and interacts with intertemporal adjustment costs.

5.3 Capacity and aggregate-pressure regimes

Figure 5.4 reports a comparative-static surface over capacity scale and aggregate-pressure strength. It is not a calibration. It summarizes the qualitative prediction of the model: lower capacity increases the shadow cost of the common resource, while aggregate-pressure costs affect the location and intensity of the binding region.

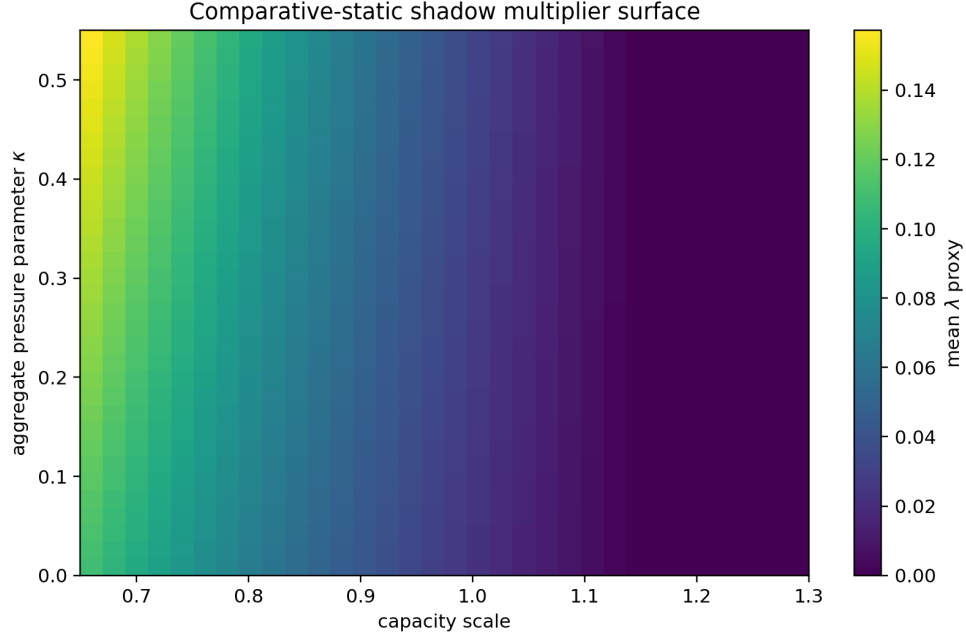


Figure 5.4: Comparative-static surface for the common multiplier across capacity and aggregate-pressure regimes. Darker regions correspond to stronger shared-capacity scarcity.

5.4 Contract-level wedges

At the peak stress date, each option contract is assigned a marginal risk-capital intensity from its delta, gamma, maturity, and vega. The model-implied premium follows the marginal-cost rule (4.13), and the implied-volatility wedge follows (4.16). Figure 5.5 shows the short-maturity implied-volatility curve before and after adding the funding wedge. The stress effect is not a uniform parallel shift; it varies across strikes because marginal risk-capital intensity varies across contracts.

Figures 5.6a and 5.6b show the cross-sectional wedge more directly. Contracts with higher marginal risk-capital intensity receive larger implied-volatility wedges per unit vega. The inventory generation is smoothed across adjacent strikes and maturities so that the figure illustrates a graduated cross-section rather than a corner solution concentrated in one contract. This is the numerical counterpart of Proposition 4.6.

6 Discussion

The paper should be read as a theory benchmark, not as an empirical claim about a specific dataset. The model does not estimate dealer inventories, infer proprietary balance-sheet constraints, or calibrate to a particular option surface.

The quadratic risk-capital constraint is also a benchmark. Real dealer balance-sheet costs may be linear in gross notional, haircut-based, VaR-based, or shaped by regulatory ratios such as leverage requirements. The quadratic formulation is used because it provides a smooth convex setting in which the variational equilibrium, the common multiplier, and the premium decomposition can be derived explicitly. A nonsmooth version with linear haircut constraints would replace gradients by subgradients and preserve the same broad equilibrium logic, but it would obscure the closed-form comparative statics.

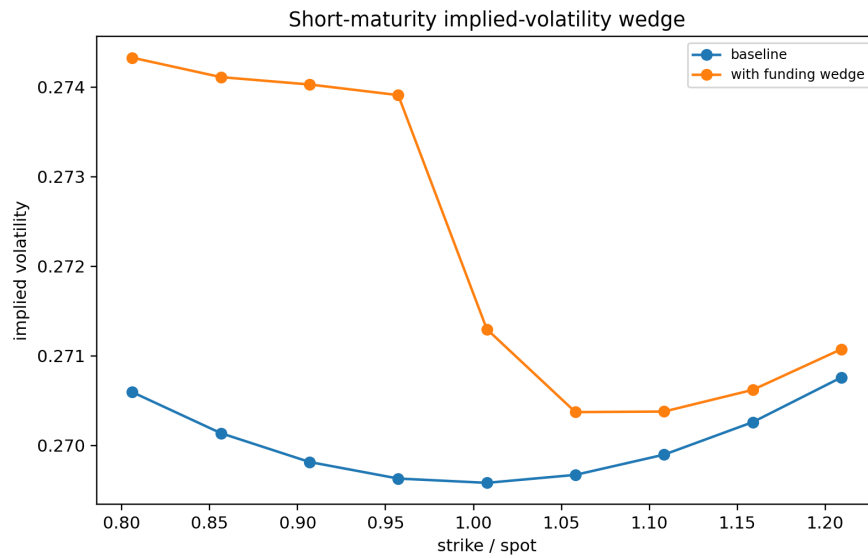


Figure 5.5: Short-maturity implied-volatility curve in the baseline state and after applying the model-implied risk-capital wedge.

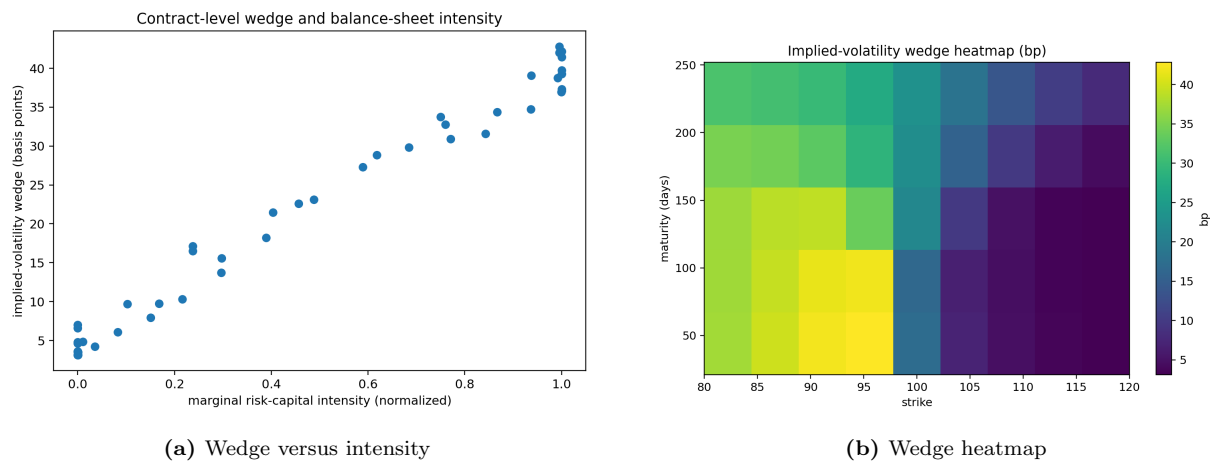


Figure 5.6: Contract-level implied-volatility wedges at peak stress. The smoothed heterogeneous inventory panel produces a graduated cross-section: balance-sheet-intensive contracts absorb larger wedges when the common multiplier is high.

The variational selection is economically restrictive. It is appropriate when dealers face a common clearing constraint, a common prime broker, a centralized funding desk, or homogeneous funding spreads. It is less appropriate when dealer funding costs are heterogeneous and bilateral. In that case, a general GNE with dealer-specific multipliers would be the natural extension. The present model studies the common-resource benchmark.

Finally, the implied-volatility wedge is not assumed directly. It is derived from a marginal dealer pricing rule and a first-order vega approximation. The model predicts cross-sectional implied-volatility wedges through marginal equilibrium cost scaled by vega. The numerical section should therefore be read as structural comparative statics of this mechanism, not as empirical validation.

7 Conclusion

This paper developed a dynamic multi-dealer theory model of option hedging under a shared risk-capital constraint. Dealers carry hedge inventory through time, face trade-adjustment costs, and interact through aggregate hedge pressure. The shared capacity constraint turns the problem into a generalized Nash game. The variational equilibrium is characterized by a finite-dimensional variational inequality and, in the quadratic benchmark, by the minimizer of an exact convex potential.

The main results show that a common multiplier prices the shared capacity constraint, tighter capacity raises that multiplier on binding branches, aggregate-pressure costs damp common-direction inventory shocks, and the optimized potential yields a contract-level premium through an envelope formula. A first-order vega expansion then maps the premium into an implied-volatility wedge. Structural numerical illustrations show the mechanism in a heterogeneous multi-dealer option book. The framework provides a tractable theoretical foundation for studying option-dealer hedging when the scarce resource is not only risk tolerance, but shared risk-capital capacity.

8 Declarations

Acknowledgments

No non-author contributions are acknowledged in this draft.

Funding

No external funding was received for this research.

Conflict of Interest

The author declares no conflict of interest.

Data Availability Statement

The numerical simulations use synthetic data generated by the reproducibility script included with this submission bundle. No proprietary or third-party empirical dataset is used.

A Proofs

Proof of Proposition 4.1. Differentiating the potential (4.1) with respect to dealer i 's trade sequence gives the hedge-error gradient, the private trading-cost gradient, the adjustment-cost gradient, and the aggregate pressure term κU_t at each date. By (3.6), this is exactly the gradient of J_i with respect to u_i . Hence $F = \nabla \mathcal{P}$. $\square$

Proof of Proposition 4.2. Since $F = \nabla \mathcal{P}$, the first-order condition for minimizing the convex function $\mathcal{P}$ over the convex set $\mathcal{K}$ is

$$\langle \nabla \mathcal{P}(u^*), u - u^* \rangle \geq 0, \quad \forall u \in \mathcal{K}.$$

This is exactly the variational inequality (3.15). Strong convexity gives uniqueness of the minimizer and hence uniqueness of the variational equilibrium. $\square$

Proof of Theorem 4.3. The feasible set is nonempty by assumption. Since $\alpha_i > 0$ and $\sum_i m_{i,t}(u_i) \leq M_t < \infty$ for each date, each $u_{i,t}$ is bounded. The feasible set is therefore closed, bounded, and convex. The potential is continuous and convex, so it attains a minimum on $\mathcal{K}$. Proposition 4.2 identifies any minimizer as a variational equilibrium. $\square$

Proof of Theorem 4.4. The term $\sum_{i,t} \eta_i u_{i,t}^2/2$ is strongly convex when $\eta_i > 0$ for all i . The remaining quadratic terms are convex under the stated nonnegativity assumptions. Hence $\mathcal{P}$ is strongly convex on $\mathcal{K}$. A strongly convex function has at most one minimizer on a convex set, so the variational equilibrium is unique. $\square$

Proof of Proposition 4.5. The stationarity equation (4.8) implies

$$(A + 2\lambda Q) \frac{du}{d\lambda} + 2(Qu(\lambda) + a) = 0.$$

Since $A \succ 0$ and $Q \succeq 0$, $A + 2\lambda Q$ is positive definite for $\lambda \geq 0$. Hence

$$\frac{du}{d\lambda} = -2(A + 2\lambda Q)^{-1}(Qu(\lambda) + a).$$

Using $\nabla q(u) = 2(Qu + a)$ gives

$$\varphi'(\lambda) = \nabla q(u(\lambda))^\top \frac{du}{d\lambda} = -4(Qu(\lambda) + a)^\top (A + 2\lambda Q)^{-1}(Qu(\lambda) + a) \leq 0.$$

The inequality is strict whenever $Qu(\lambda) + a \neq 0$. Since the binding branch satisfies $\varphi(\lambda) = M$, a lower value of M requires a weakly larger value of λ , and strictly so when the constraint has nonzero local marginal exposure. $\square$

Derivation of (4.11). This derivation is for the homogeneous one-date special case described in Section 4.5. The matrix $A = (\rho + \eta)I + \kappa \mathbf{1}\mathbf{1}^\top$ has eigenvalue $\rho + \eta + \kappa N$ in the direction of $\mathbf{1}$ and eigenvalue $\rho + \eta$ on $\{\mathbf{1}\}^\perp$. The unconstrained one-date response is $u = \rho A^{-1}d$. Decomposing $d = d^\parallel + d^\perp$ into the two invariant subspaces gives (4.11). This argument is not intended to cover the heterogeneous finite-horizon model with dealer-specific coefficients and dynamic adjustment costs; there the response operator is the inverse of the full dynamic Hessian. $\square$

Proof of Proposition 4.6. The value function (4.12) is that of a parametric convex program. Under differentiability and strict complementarity, the active set is locally stable and the derivative of the optimized value with respect to q_ℓ is obtained by differentiating the Lagrangian at the optimizer. This gives (4.14). The implied-volatility expression follows from the first-order Taylor expansion (4.15) and the assumption $\mathcal{V}_\ell > 0$. $\square$

B Normal distribution notation

The standard normal density and distribution used in the option-Greek construction are

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad N(x) = \int_{-\infty}^x \varphi(y) dy.$$

References

- [1] Adrian, T., Etula, E., & Muir, T. (2014). Financial intermediaries and the cross-section of asset returns. *Journal of Finance*, 69(6), 2557–2596.

- [2] Avellaneda, M., & Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3), 217–224.
- [3] Bollen, N. P. B., & Whaley, R. E. (2004). Does net buying pressure affect the shape of implied volatility functions? *Journal of Finance*, 59(2), 711–753.
- [4] Bouchard, B., Loeper, G., & Zou, Y. (2017). Hedging of covered options with linear market impact and gamma constraint. *SIAM Journal on Control and Optimization*, 55(5), 3319–3348.
- [5] Brunnermeier, M. K., & Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies*, 22(6), 2201–2238.
- [6] Buis, B., Pieterse-Bloem, M., Verschoor, W. F. C., & Zwinkels, R. C. J. (2024). Gamma positioning and market quality. *Journal of Economic Dynamics and Control*, 164, 104880.
- [7] Carlin, B. I., Lobo, M. S., & Viswanathan, S. (2007). Episodic liquidity crises: Cooperative and predatory trading. *Journal of Finance*, 62(5), 2235–2274.
- [8] Cetin, U., Jarrow, R. A., & Protter, P. (2004). Liquidity risk and arbitrage pricing theory. *Finance and Stochastics*, 8(3), 311–341.
- [9] Facchinei, F., & Pang, J.-S. (2003). *Finite-dimensional variational inequalities and complementarity problems*. New York, NY: Springer.
- [10] Facchinei, F., Fischer, A., & Piccialli, V. (2007). On generalized Nash games and variational inequalities. *Operations Research Letters*, 35(2), 159–164.
- [11] Facchinei, F., & Kanzow, C. (2010). Generalized Nash equilibrium problems. *4OR*, 8(3), 173–210.
- [12] Frey, R. (1998). Perfect option hedging for a large trader. *Finance and Stochastics*, 2(2), 115–141.
- [13] Frey, R., & Stremme, A. (1997). Market volatility and feedback effects from dynamic hedging. *Mathematical Finance*, 7(4), 351–374.
- [14] Garleanu, N., & Pedersen, L. H. (2011). Margin-based asset pricing and deviations from the law of one price. *Review of Financial Studies*, 24(6), 1980–2022.
- [15] Garleanu, N., Pedersen, L. H., & Poteshman, A. M. (2009). Demand-based option pricing. *Review of Financial Studies*, 22(10), 4259–4299.
- [16] Golez, B., Jackwerth, J. C., Slavutskaya, A., & Vashchilko, A. (2024). Funding illiquidity implied by S&P 500 derivatives. *Risks*, 12(9), 149.
- [17] Gueant, O., Lehalle, C.-A., & Fernandez-Tapia, J. (2013). Dealing with the inventory risk: A solution to the market making problem. *Mathematics and Financial Economics*, 7(4), 477–507.
- [18] Harker, P. T. (1991). Generalized Nash games and quasi-variational inequalities. *European Journal of Operational Research*, 54(1), 81–94.
- [19] He, Z., & Krishnamurthy, A. (2013). Intermediary asset pricing. *American Economic Review*, 103(2), 732–770.
- [20] Kulkarni, A. A., & Shanbhag, U. V. (2012). On the variational equilibrium as a refinement of the generalized Nash equilibrium. *Automatica*, 48(1), 45–55.
- [21] Lakonishok, J., Lee, I., Pearson, N. D., & Poteshman, A. M. (2007). Option market activity. *Review of Financial Studies*, 20(3), 813–857.
- [22] Ni, S. X., Pearson, N. D., & Poteshman, A. M. (2005). Stock price clustering on option expiration dates. *Journal of Financial Economics*, 78(1), 49–87.

- [23] Nemirovski, A. (2004). Prox-method with rate of convergence $O(1/t)$ for variational inequalities with Lipschitz continuous monotone operators and smooth convex-concave saddle point problems. *SIAM Journal on Optimization*, 15(1), 229–251.
- [24] Rosen, J. B. (1965). Existence and uniqueness of equilibrium points for concave N -person games. *Econometrica*, 33(3), 520–534.
- [25] Schied, A., & Zhang, T. (2019). A market impact game under transient price impact. *Mathematics of Operations Research*, 44(1), 102–121.
- [26] Sircar, K. R., & Papanicolaou, G. (1998). General Black-Scholes models accounting for increased market volatility from hedging strategies. *Applied Mathematical Finance*, 5(1), 45–82.
- [27] Soner, H. M., & Touzi, N. (2000). Superreplication under gamma constraints. *SIAM Journal on Control and Optimization*, 39(1), 73–96.
- [28] Stoikov, S., & Saglam, M. (2009). Option market making under inventory risk. *Review of Derivatives Research*, 12(1), 55–79.